Question	Answer	Marks	Partial Marks
1(a)	Image at (4, –1) (4, –4) (5, –4)	**2**	**B1** for translation by $\begin{pmatrix} 8 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -6 \end{pmatrix}$ or for correct vertices not joined
1(b)	Image at (–4, –4) (–4, –7) (–3, –4)	**2**	**B1** for reflection in $x = -1$ or $y = k$ or for correct vertices not joined
1(c)	Enlargement 3 (–5, 5)	**3**	**B1** for each
1(d)	Rotation 90º clockwise oe (1, 1)	**3**	**B1** for each
2(a)	1 : 5 : 12	**2**	**M1** for 2 : 10 : 24 or 7 : 35 : 84 or $\frac{1}{18} : \frac{5}{18} : \frac{12}{18}$
2(b)(i)	266 and 95	**3**	**B2** for 266 or 95 or 266 and 95 reversed or **M1** for $\frac{114}{6}$
2(b)(ii)	15	**2**	**M1** for $\frac{114 - 96.9}{114}$ [× 100] oe or $\frac{96.9}{114} \times 100$
2(c)(i)	2h 50min	**1**	
2(c)(ii)	636	**2**	**M1** for 1802 ÷ *their* 2h 50min
3(a)	Disagree: the median for the women is greater (than the median for the men) oe Disagree: the men have a smaller [interquartile] range of times oe	**2**	**B1** for each correct statement oe
3(b)(i)	87.4 nfww	**4**	**M1** for mid-points **soi** (30, 80, 130, 190, 270) **M1** for use of Σ*fm* with *m* in correct interval including both boundaries **M1** (dep on 2nd M1) for Σ*fm* ÷ (41 + 24 + 23 + 8 + 4)
3(b)(ii)(a)	90	**1**	
3(b)(ii)(b)	8	**2**	**B1** for 92 seen

Question	Answer	Marks	Partial Marks
3(b)(iii)	2.4	**2**	**M1** for $\frac{24}{40}$ or $\frac{8}{60}$ Or **B1** for [multiplier] 18 or $\frac{1}{18}$
4(a)	38.6	**3**	**M2** for [2 ×] (8.5 + 0.05 + 10.7 + 0.05) or **M1** for 8.5 + 0.05 or 10.7 + 0.05
4(b)(i)	8.86 or 8.863…	**2**	**M1** for $\frac{h}{9}$ = sin 80 or better oe
4(b)(ii)	∠CDF = 100 leading to ∠*DCF* = 40 Or ∠EDF = 80 leading to ∠*DCF* = 40	**M1**	Implied by 180-(100 + 40) = 40 or 80 – 40
	'two equal angles'	**A1**	With no incorrect work seen
4(b)(iii)	66.5 or 66.45 to 66.47…	**3**	**M2** for 0.5(3 + 12) × *their* **(b)(i)** or 12 × *their* **(b)(i)** – 0.5 × 9 × 9 × sin 100 oe or **B1** for *DC* = 9 or *BC* = 3